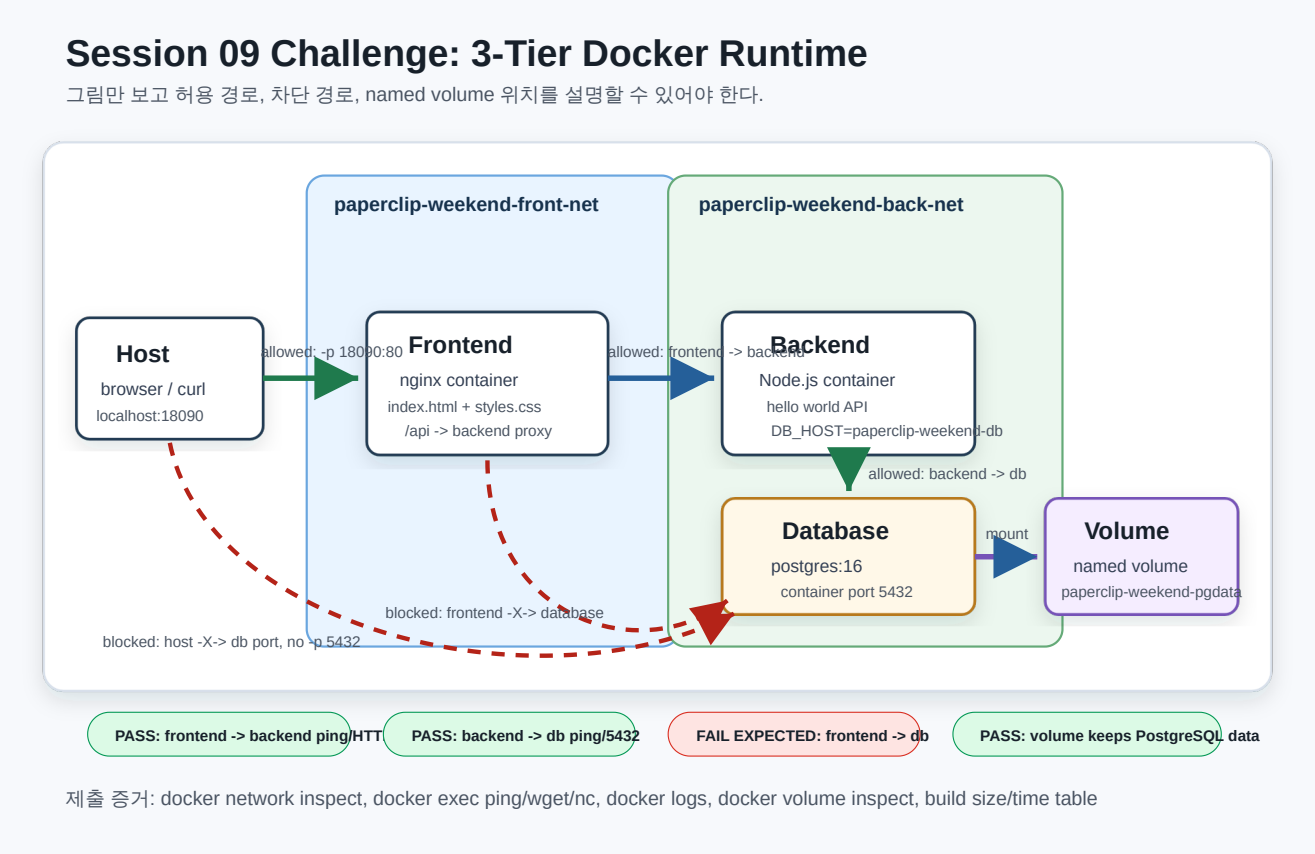
Session 09 Challenge: 3-Tier Image Delivery and Runtime Evidence

이 주말 챌린지는 Day 3의 image/build/tag 기준을 3-tier 구조로 확장한다. 목표는 멋진 기능을 많이 만드는 것이 아니라, frontend, backend, database를 분리해 실행하고 **네트워크, 볼륨, 로그, 트러블슈팅, 이미지 최적화 수치**를 증거로 남기는 것이다.

성공 기준은 의도적으로 단순하다. frontend는 `index.html`과 `styles.css`를 nginx로 제공한다. backend는 Node.js hello world API를 띄운다. database는 `postgres:16` container와 named volume이 정상 생성되고 유지되는 것만 확인해도 성공이다.

Architecture



Tier	Container	Image	Network	역할
Frontend	`paperclip-weekend-front`	직접 build	`paperclip-weekend-front-net`	`index.html`, `styles.css` 제공, backend로 proxy
Backend	`paperclip-weekend-back`	직접 build	`paperclip-weekend-front-net`, `paperclip-weekend-back-net`	Node.js hello world API, DB network 확인
Database	`paperclip-weekend-db`	`postgres:16`	`paperclip-weekend-back-net`	PostgreSQL 16, named volume 사용

그림 해석 기준:

- Host는 frontend의 `18090:80` publish port로만 들어간다.
- Frontend는 `paperclip-weekend-front-net`에만 붙고 backend만 호출한다.
- Backend는 `front-net`, `back-net` 양쪽에 붙는 유일한 application tier다.
- Database는 `paperclip-weekend-back-net`에만 붙고 host port를 publish하지 않는다.
- PostgreSQL data는 container filesystem이 아니라 `paperclip-weekend-pgdata` named volume에 남는다.

Network rule:

```
host -> frontend
frontend -> backend
backend -> database
frontend -X-> database
host -X-> database port
```

`frontend`는 `backend`와 같은 network에만 붙인다. `database`는 `backend`와 같은 network에만 붙인다. 따라서 frontend에서 database 이름을 바로 찾을 수 없어야 한다. `ping`만 성공해도 연결 증거로 인정하지만, 가능하면 HTTP 또는 port check도 함께 남긴다.

File Map

Path	역할
`frontend/Dockerfile`	frontend nginx image
`assets/session-09-3tier-architecture.svg`	network, connection rule, volume 구조도
`frontend/Dockerfile.size-compare`	frontend base image size 비교
`frontend/.dockerignore`	frontend build context에서 `.env`, `node_modules`, output 제외
`frontend/nginx.conf`	`/api/` 요청을 backend로 proxy
`frontend/html/index.html`	browser 확인용 정적 HTML
`frontend/html/styles.css`	browser 확인용 정적 CSS
`backend/Dockerfile`	backend Node.js hello world image
`backend/Dockerfile.size-compare`	backend base image size 비교
`backend/.dockerignore`	backend build context에서 `node_modules`, cache, secret 제외
`backend/server.js`	`/health`, `/api/info` 응답
`scripts/measure-build.sh`	최초 build 시간과 image size 측정
`SUBMISSION.md`	제출 보고서 템플릿

Lab root:

```
cd /mnt/d/paperclip/week2/day3/labs/weekend-3tier-challenge
```

1. Build

```
cd /mnt/d/paperclip/week2/day3/labs/weekend-3tier-challenge
docker build -t paperclip-weekend-frontend:optimized ./frontend
docker build -t paperclip-weekend-backend:optimized ./backend
docker images 'paperclip-weekend-*'
```

Expected:

```
paperclip-weekend-frontend   optimized
paperclip-weekend-backend    optimized
```

2. Create network and volume

```
docker network create paperclip-weekend-front-net || true
docker network create paperclip-weekend-back-net || true
docker volume create paperclip-weekend-pgdata
docker network ls | grep paperclip-weekend
docker volume ls | grep paperclip-weekend
```

Expected:

```
paperclip-weekend-front-net
paperclip-weekend-back-net
paperclip-weekend-pgdata
```

3. Run DB

```
docker rm -f paperclip-weekend-db || true
docker run -d \
  --name paperclip-weekend-db \
  --network paperclip-weekend-back-net \
  -e POSTGRES_PASSWORD=weekend-only \
  -e POSTGRES_DB=weekend \
  -v paperclip-weekend-pgdata:/var/lib/postgresql/data \
  postgres:16

docker logs paperclip-weekend-db --tail 40
```

Expected:

```
database system is ready to accept connections
```

4. Run backend

```
docker rm -f paperclip-weekend-back || true
docker run -d \
  --name paperclip-weekend-back \
  --network paperclip-weekend-back-net \
  -e APP_ENV=weekend \
  -e DB_HOST=paperclip-weekend-db \
```

```
  -e DB_PORT=5432 \
  paperclip-weekend-backend:optimized

docker network connect paperclip-weekend-front-net paperclip-weekend-back
docker logs paperclip-weekend-back --tail 40
```

Expected:

```
node backend listening on 8080
```

5. Run frontend

```
docker rm -f paperclip-weekend-front || true
docker run -d \
  --name paperclip-weekend-front \
  --network paperclip-weekend-front-net \
  -p 18090:80 \
  paperclip-weekend-frontend:optimized

curl -I http://localhost:18090
curl -s http://localhost:18090/api/info
```

Expected:

```
HTTP/1.1 200 OK
"service": "node-backend"
```

6. Network evidence

Frontend to backend must work. `ping` 성공만으로도 network 연결 증거로 인정하고, HTTP는 application path 확인으로 추가한다.

```
docker exec paperclip-weekend-front ping -c 2 paperclip-weekend-back
docker exec paperclip-weekend-front wget -q -O- http://paperclip-weekend-
back:8080/health
```

Expected:

```
2 packets transmitted
ok
```

Backend to DB must work. ping만 성공해도 인정한다.

```
docker exec paperclip-weekend-back ping -c 2 paperclip-weekend-db
docker exec paperclip-weekend-back sh -c 'nc -z paperclip-weekend-db 5432 &&
echo db-port-open'
```

Expected:

```
2 packets transmitted
db-port-open
```

Frontend to DB must fail:

```
docker exec paperclip-weekend-front ping -c 2 paperclip-weekend-db || true
docker exec paperclip-weekend-front wget -q -O- http://paperclip-weekend-
db:5432 || true
```

Expected failure:

```
bad address 'paperclip-weekend-db'
```

Interpretation:

```
frontend는 front-net에만 있다.
db는 back-net에만 있다.
backend만 두 network에 모두 붙어 있으므로 frontend -> backend, backend -> db 흐름
만 허용된다.
```

7. Volume evidence

```
docker volume inspect paperclip-weekend-pgdata
docker logs paperclip-weekend-db --tail 20
```

Expected:

```
"Name": "paperclip-weekend-pgdata"
database system is ready to accept connections
```

추가 확인:

```
docker rm -f paperclip-weekend-db
docker run -d \
  --name paperclip-weekend-db \
  --network paperclip-weekend-back-net \
  -e POSTGRES_PASSWORD=weekend-only \
  -e POSTGRES_DB=weekend_changed \
  -v paperclip-weekend-pgdata:/var/lib/postgresql/data \
  postgres:16
docker logs paperclip-weekend-db --tail 40
```

Expected signal:

```
Database directory appears to contain a database; Skipping initialization
```

Interpretation:

```
named volume에 기존 DB data가 남아 있으므로 POSTGRES_DB를 바꿔도 최초 초기화가 다시 실행되지 않는다.
```

8. Logs and troubleshooting evidence

```
docker logs paperclip-weekend-front --tail 40
docker logs paperclip-weekend-back --tail 40
docker logs paperclip-weekend-db --tail 40
docker ps --filter name=paperclip-weekend
docker inspect paperclip-weekend-back --format '{{json
.NetworkSettings.Networks}}'
```

RCA mini drill:

Symptom	First evidence command	Likely cause	Fix
frontend `/api/info` returns 502	`docker logs paperclip-weekend-front`	backend not on front network	`docker network connect paperclip-weekend-front-net paperclip-weekend-back`
backend cannot reach DB	`docker exec paperclip-weekend-back ping ...`	backend not on back network or DB name wrong	connect backend to back network, check container name
DB data does not reset	`docker logs paperclip-weekend-db`	stale named volume	decide whether to remove volume
build is slow or image is large	`scripts/measure-build.sh`	base image or context size	compare alpine/slim/default, check `.dockerignore`

9. Build speed and image size measurement

최초 build 속도와 image size를 반드시 숫자로 남긴다. cache가 섞이면 수치가 흐려지므로 `--no-cache`로 측정한다. 처음 실행하는 machine에서는 base image pull 시간이 포함될 수 있다. 그래서 제출할 때는 처음 `pull` 포함, 이미 `pull`된 상태 중 어느 조건인지 함께 적는다.

Build context hygiene check:

```
sed -n '1,120p' frontend/.dockerignore
sed -n '1,120p' backend/.dockerignore
du -sh frontend backend
```

Expected:

```
.env, .env.*, node_modules, dist, build, __pycache__, .venv 같은 파일/디렉터리가 제외 규칙에 있다.
```

```
chmod +x scripts/measure-build.sh
./scripts/measure-build.sh
```

Expected output:

```
target,base,seconds,size_bytes,size_human
frontend,nginx:stable,...
frontend,nginx:stable-alpine,...
frontend,nginx:stable-trixie,...
backend,node:22,...
```

```
backend,node:22-slim,...
backend,node:22-alpine,...
```

제출 표에는 다음을 기록한다.

Target	Base image	First build seconds	Size	선택 여부	이유
frontend	`nginx:stable`				
frontend	`nginx:stable-alpine`				
frontend	`nginx:stable-trixie`				
backend	`node:22`				
backend	`node:22-slim`				
backend	`node:22-alpine`				

Image size가 커지면 push/pull 시간, registry storage, runner/node pull time, 취약점 scan 시간이 늘어날 수 있다. 가장 작은 image가 항상 정답은 아니지만, 선택 근거는 숫자로 설명해야 한다.

10. Upload or submission

Docker Hub push는 선택이다. push한다면 public/private 범위와 secret 포함 여부를 먼저 확인한다.

```
docker tag paperclip-weekend-frontend:optimized <dockerhub-id>/paperclip-weekend-frontend:week2-day3
docker tag paperclip-weekend-backend:optimized <dockerhub-id>/paperclip-weekend-backend:week2-day3
# docker push <dockerhub-id>/paperclip-weekend-frontend:week2-day3
# docker push <dockerhub-id>/paperclip-weekend-backend:week2-day3
```

Push하지 않는 경우에도 `SUBMISSION.md`를 작성해 제출한다.

Required evidence:

```
1. architecture table
2. network evidence: frontend -> backend success, backend -> db success, frontend -> db fail
3. volume evidence: pgdata inspect and stale volume interpretation
4. logs: frontend/backend/db key lines
5. troubleshooting RCA 1개 이상
6. build speed and image size table
7. image tag or optional registry URL
```

Cleanup

```
docker rm -f paperclip-weekend-front paperclip-weekend-back paperclip-weekend-db || true
docker network rm paperclip-weekend-front-net paperclip-weekend-back-net || true

# data reset이 필요할 때만 실행한다.
# docker volume rm paperclip-weekend-pgdata

# image까지 지울 때만 실행한다.
# docker image rm paperclip-weekend-frontend:optimized paperclip-weekend-backend:optimized
```